

AER626 Finite Element Analysis of an Inverted T-Stringer Aircraft Component

Project Overview

In Winter 2026, for AER626 Applied Finite Elements, I analysed an aircraft structural component made of three bonded members: a stainless steel 304 flat plate, an Al 6061-T6 angle stringer and an Al 7075-T6 angle stringer, assembled into an inverted T. The assembly was modelled in SolidWorks, exported as a STEP file and imported into ANSYS Mechanical APDL with each volume kept separate so materials could be assigned individually.

Two analyses were run: a static analysis for von Mises stress and displacement under a parabolic in-plane tensile load, and a free vibration analysis for the first three natural frequencies and mode shapes. Both were repeated for two load orientations, three mesh densities and two element types to check convergence and see how sensitive the results were to modelling choices.

Model Setup

- **Bonding.** All contact surfaces were joined with the Glue Boolean operation so the volumes share nodes at the interfaces, matching the specification that the rivets act as full attachments.
- **Materials.** SS 304 ($E = 28.0 \times 10^6$ psi, $\nu = 0.29$), Al 6061-T6 ($E = 10.0 \times 10^6$ psi, $\nu = 0.33$) and Al 7075-T6 ($E = 10.4 \times 10^6$ psi, $\nu = 0.33$), all linear elastic and isotropic.
- **Elements.** SOLID185, an 8-node brick, and SOLID186, a 20-node brick with mid-side nodes that handles bending and stress gradients better at the same mesh density.
- **Meshes.** Coarse M1 (0.20 in), medium M2 (0.15 in) and fine M3 (0.075 in).
- **Loading.** The parabolic load $P(y) = 20000(5y - y^2)/6.25$ lb/in was applied as individual nodal forces along the loaded edge, computed at each node's y-coordinate and scaled by its tributary length, rather than through the ANSYS function file tool.
- **Orientations.** Design 1 fixed edge ab with rollers on cd and load along cd. Design 2 swapped to the perpendicular sides: ac fixed, rollers on bd, load along bd.

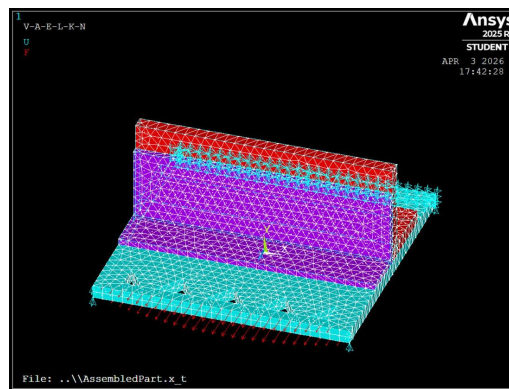


Figure 1. Design 1 mesh with boundary conditions and parabolic nodal loads applied in ANSYS.

Results

Design 1. Peak von Mises stress fell from 374,602 psi on the coarse mesh to 217,841 psi on the medium mesh (a 41.8 percent drop) and then to 216,556 psi on the fine mesh, a change of only 0.6 percent. Maximum displacement settled between 0.0088 and 0.0094 in over the last two meshes. The static solution is well converged at M3, and the coarse mesh had overpredicted the peak stress by more than 40 percent.

Design 2. Peak stress went 84,407 psi, 76,470 psi, then 130,709 psi, a 70.9 percent jump from M2 to M3. The finest mesh resolved a stress concentration at the fixed-edge corner that the coarser meshes had smeared out, so Design 2 is not converged even at M3 and would need local refinement at that corner to confirm the peak.

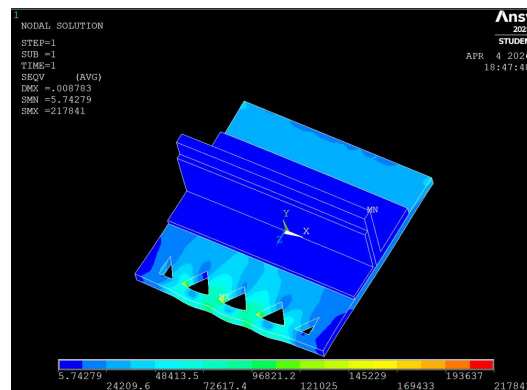


Figure 2. Von Mises stress for Design 1 on the medium mesh, SOLID185. Stress peaks at the stringer-plate interface and the fixed-edge corners.

Natural frequencies. Using the Block Lanczos solver, Design 1 gave first-mode frequencies of 79.8, 69.3 and 53.6 Hz on M1, M2 and M3, with all three modes dropping 19 to 23 percent between the last two meshes. Design 2 behaved the same way for modes 1 and 2 (57.1 and 105.8 Hz at M3) while mode 3 was more stable at 7.4 percent change. Coarser meshes overestimate stiffness and therefore frequency, and the dynamic solution was still converging at M3 even though the mode shapes themselves were stable.

Element type. Repeating the coarse-mesh runs with SOLID186 gave results closer to the fine SOLID185 mesh, which is the expected benefit of mid-side nodes in bending-dominated regions, at the cost of more degrees of freedom per element.

Discussion

The most useful outcome of the project was seeing how different the two orientations behaved under the same refinement. Design 1 converged cleanly; Design 2 looked converged at M2 and was not. A single mesh, or even two, would have given a wrong answer with no warning. Von Mises stress was chosen as the reporting quantity because it gives one scalar to compare against yield for a multi-material assembly, and the peak locations at the interfaces and fixed corners matched where the load path turns and reactions are highest.

Skills and Takeaways

- Technical: ANSYS Mechanical APDL (geometry import, glue bonding, material and element assignment, manual nodal loading, static and modal solutions), mesh convergence studies, SOLID185 versus SOLID186 comparison, and SolidWorks assembly modelling.
 - Professional: planning a study with enough cases to detect a non-converged result, and reporting a result that did not converge rather than presenting the fine-mesh number as final.
-

Supplementary materials. The full technical report, with all displacement, von Mises and mode shape plots for both designs, every mesh and both element types, plus the SolidWorks assembly drawing, is available here: [AER626 StressAnalysis FullReport](#). This work was completed individually by **Gaber Soltan** for AER626 at **Toronto Metropolitan University** (Winter 2026).